EASY · ALGEBRA · ANSWER KEY

Algebra

19

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

D

D.

x	y
0	-5
1	-3
2	-1

Choice D is correct. It's given that the graph shows the linear relationship between x and y. The given graph passes through the points $0, -5$, $1, -3$, and $2, -1$. It follows that when $x = 0$, the corresponding value of y is -5, when $x = 1$, the corresponding value of y is -3, and when $x = 2$, the corresponding value of y is -1. Of the given choices, only the table in choice D gives these three values of x and their corresponding values of y for the relationship shown in the graph.

Choice A is incorrect. This table represents a relationship between x and y such that the graph passes through the points $0, 0$, $1, -7$, and $2, -9$.

Choice B is incorrect. This table represents a relationship between x and y such that the graph passes through the points $0, 0$, $1, -3$, and $2, -1$.

Choice C is incorrect. This table represents a linear relationship between x and y such that the graph passes through the points $0, -5$, $1, -7$, and $2, -9$.

Q2

D

D. 3

Choice D is correct. Subtracting $16x$ from both sides of the given equation yields $24 = 8x$. Dividing both sides of this equation by 8 yields $3 = x$. Therefore, the solution to the given equation is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3

D

D. $y = 4.8x + 162$

Choice D is correct. It's given that the length of the whale was 162 cm when it was born and that its length increased an average of 4.8 cm per month for the first 12 months after it was born. Since x represents the number of months after the whale was born, the total increase in the whale's length, in cm, is 4.8 times x, or $4.8x$. The length of the whale y, in cm, can be found by adding the whale's length at birth, 162 cm, to the total increase in length, $4.8x$ cm. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q4 A

A. $y = 6x + 5$

Choice A is correct. An equation that defines a line in the xy-plane can be written as $y = mx + b$, where m is the slope of the line and $(0, b)$ is its y-intercept. It's given that line k is defined by the equation $y = 6x + 4$; therefore, the slope of line k is 6. Since line j is parallel to line k in the xy-plane and parallel lines have equal slopes, it follows that the slope of line j is 6. It's also given that line j passes through the point $(0, 5)$, which is its y-intercept. Substituting 6 for m and 5 for b in the equation $y = mx + b$ yields $y = 6x + 5$. Therefore, the equation $y = 6x + 5$ defines line j.

Choice B is incorrect. This equation defines a line that has a slope of -5, not 6.

Choice C is incorrect. This equation defines a line that has a slope of -6, not 6.

Choice D is incorrect. This equation defines a line that has a slope of 5, not 6.

Q5 B

B. 350

Choice B is correct. Subtracting 7 from each side of the given equation yields $w = 350$. Therefore, the value of w that is the solution to the given equation is 350.

Choice A is incorrect. This is the value of w that is the solution to the equation $7w = 357$, not $w + 7 = 357$.

Choice C is incorrect. This is the value of w that is the solution to the equation $w - 7 = 357$, not $w + 7 = 357$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

A

A. 0

Choice A is correct. For the linear function f, it's given that $f7 = 28$. Substituting 7 for x and 28 for fx in the given function yields $28 = 47 + b$, or $28 = 28 + b$. Subtracting 28 from each side of this equation yields $0 = b$. Therefore, the value of b is 0.

Choice B is incorrect. Substituting 1 for b in the given function yields $fx = 4x + 1$. For this function, when the value of x is 7, the value of fx is 29, not 28.

Choice C is incorrect. Substituting 4 for b in the given function yields $fx = 4x + 4$. For this function, when the value of x is 7, the value of fx is 32, not 28.

Choice D is incorrect. Substituting 7 for b in the given function yields $fx = 4x + 7$. For this function, when the value of x is 7, the value of fx is 35, not 28.

Q7

A

A.

x	0	4	8
y	21	28	35

Choice A is correct. To verify which table represents this linear relationship, the values in each table can be checked by substituting them into the given equation. The table in choice A shows that when $x = 0$, $y = 21$. Substituting these values into the given equation yields $7(0) - 4(21) = -84$, or $-84 = -84$, which is true. Additionally, the table in choice A shows that when $x = 4$, $y = 28$. Substituting these values into the given equation yields $7(4) - 4(28) = -84$, or $-84 = -84$, which is true. Finally, the table in choice A shows that when $x = 8$, $y = 35$. Substituting these values into the given equation yields $7(8) - 4(35) = -84$, or $-84 = -84$, which is true. Therefore, the table in choice A gives three values of x and their corresponding values of y.

Choice B is incorrect. The table in choice B shows that when $x = 0$, $y = 35$. Substituting these values into the given equation yields $7(0) - 4(35) = -84$, or $-140 = -84$, which is not true.

Choice C is incorrect. The table in choice C shows that when $x = 21$, $y = 0$. Substituting these values into the given equation yields $7(21) - 4(0) = -84$, or $147 = -84$, which is not true.

Choice D is incorrect. The table in choice D shows that when $x = 21$, $y = 8$. Substituting these values into the given equation yields $7(21) - 4(8) = -84$, or $115 = -84$, which is not true.

Q8 –

Answer not provided in source data — see rationale below.

Choice B is correct. Subtracting the number of books left over from the total number of books results in $43-7=36$, which is the number of books distributed. Dividing the number of books distributed by the number of books given to each child results in $\frac{36}{2}=18$.

Choice A is incorrect and results from dividing the total number of books by the number of books given to each child, $\frac{43}{2}\approx 22$, then subtracting the number of books left over from the result, $22-7=15$. Choice C is incorrect and results from adding the number of books left over to the total number of books, $43+7=50$, then dividing the result by the number of books given to each child, $\frac{50}{2}=25$. Choice D is incorrect and results from dividing the total number of books by the number of books given to each child, $\frac{43}{2}\approx 22$, then adding the number of books left over, $22+7=29$.

Q9 C

C. 11

Choice C is correct. Substituting 4 for x in $g(x) = 5x + a$ gives $g(4) = 5(4) + a$. Since $g(4) = 31$, the equation $g(4) = 5(4) + a$ simplifies to $31 = 20 + a$. It follows that $a = 11$.

Choices A, B, and D are incorrect and may result from arithmetic errors.

Q10 A

A. 5.48

Choice A is correct. It's given that the store's sales of a certain Greek yogurt totaled $1{,}277.94$ dollars last month. It's also given that the equation $5.48x + 7.30y = 1{,}277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. Since x represents the number of smaller containers of yogurt sold, the expression $5.48x$ represents the total sales, in dollars, from smaller containers of yogurt. This means that x smaller containers of yogurt were sold at a price of 5.48 dollars each. Therefore, according to the equation, 5.48 represents the price, in dollars, of each smaller container.

Choice B is incorrect. This expression represents the total sales, in dollars, from selling y larger containers of yogurt.

Choice C is incorrect. This value represents the price, in dollars, of each larger container of yogurt.

Choice D is incorrect. This expression represents the total sales, in dollars, from selling x smaller containers of yogurt.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q8 — the source CB data omitted a structured key for these items, so the answer slot shows –. The worked solution in the rationale below each entry contains the correct value.

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